

Machine Learning Viva Questions & Answers

1. Basic Data Types

- Integer (int): Whole numbers without decimal. Example: 5, -10, 100
- Float: Decimal numbers. Example: 3.14, 2.5
- String (str): Collection of characters/text. Example: "Hello"

2. Advanced Data Types

- Image: Data in the form of pictures/photos.
- Audio: Sound or voice data.
- Video: Moving visual data with audio.

3. Confusion Matrix

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1 Score = $2 \times (Precision \times Recall) / (Precision + Recall)$

4. Naive Bayes Algorithm

Bayes Theorem:

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

- Supervised learning algorithm used for classification.
- Assumes all features are independent.
- Used in spam detection and text classification.

Advantages:

- Fast and simple
- Works well with text data

Disadvantages:

- Assumes independence of features

5. Types of Machine Learning

Supervised Learning

- Uses labeled data
- Input and output both known
- Examples: Classification, Regression

Unsupervised Learning

- Uses unlabeled data
- Finds hidden patterns
- Example: Clustering

Difference:

- Supervised uses labeled data, unsupervised uses unlabeled data.
- Supervised predicts outputs, unsupervised finds patterns.

6. Dataset Division

- Usually 70% training and 30% testing.
- Sometimes 80% training and 20% testing.

Training Data: Used to train the model.

Testing Data: Used to evaluate the model.

Extra Viva Questions

Q1. What is Machine Learning?

Machine Learning is a branch of AI where systems learn from data automatically.

Q2. What is AI?

Artificial Intelligence is the simulation of human intelligence by machines.

Q3. What is a Dataset?

A dataset is a collection of related data used for training and testing.

Q4. What is Classification?

Classification predicts categories/classes from data.

Example: Spam or Not Spam.

Q5. What is Regression?

Regression predicts continuous values like house prices.

Q6. What is Overfitting?

When a model learns training data too well and performs poorly on new data.

Q7. What is Underfitting?

When a model cannot learn patterns properly from training data.

Q8. What is Feature in ML?

A feature is an input variable used for prediction.

Q9. What is Training Data?

Data used to teach the model.

Q10. What is Testing Data?

Data used to evaluate the trained model.

Q11. Applications of Machine Learning

- Recommendation systems
- Face recognition
- Spam detection
- Self-driving cars
- Medical diagnosis

Q12. What is Deep Learning?

A subset of ML using neural networks with many layers.

Q13. What is NLP?

Natural Language Processing helps computers understand human language.

Q14. What is Computer Vision?

Field of AI that enables computers to understand images and videos.

Q15. Difference Between AI and ML

AI is the broader concept of intelligent systems.

ML is a subset of AI that learns from data.